

COURSE CODE/TITLE:	CPE212/Automating Server Management	SECTION:	CPE31S2
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ACTIVITY NO. 8	Install, Configure, and Manage Availability Monitoring tools
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1. PROCEDURE OUTPUT

1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
3. Show an output of the installed Nagios for both Ubuntu and CentOS.
4. Make sure to create a new repository in GitHub for this activity

STEP 1 Install Ansible on the control node

If your nodes still don't have Ansible, install AnsibleRun `'sudo apt update'` then `'sudo apt install -y ansible'` on the control node. This puts Ansible on the machine that will push out the setup the managed nodes don't need Ansible installed on them.

```
javillonar@wdjavillonar:~$ sudo apt update
[sudo] password for javillonar:
Hit:1 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,264 kB]
Get:6 http://ph.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [292 kB]
Get:7 http://ph.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:8 http://ph.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1,550 kB]

javillonar@wdjavillonar:~$ sudo apt install -y ansible
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ansible is already the newest version (9.2.0+dfsg-0ubuntu5).
The following package was automatically installed and is no longer required:
  libfwupd2
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 139 not upgraded.
```

```
javillonar@wdjavillonar:~$ ansible --version
ansible [core 2.16.3]
  config file = None
  configured module search path = ['/home/javillonar/.ansible/plugins/modules',
'/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/javillonar/.ansible/collections:/usr/share
/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.12.3 (main, Jun 19 2026, 12:46:00) [GCC 13.3.0] (/usr/bin/p
ython3)
  jinja version = 3.1.2
  libyaml = True
```

STEP 2 Give the control node SSH access to each server

Set up SSH access Run 'ssh-keygen -t rsa -b 4096' once, then 'ssh-copy-id user@ip' for each server (s1javi, s2javi, and later centjavi). This lets the control node log into each server without a password, which Ansible needs to work.

```
javillonar@wdjavillonar:~/CPE212_Javillonar$ ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/home/javillonar/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/javillonar/.ssh/id_rsa
Your public key has been saved in /home/javillonar/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:ouT9SN5R9clLWtEICmj7wKdwguIGIsZyBrMoXZk/eXA javillonar@wdjavillonar
The key's randomart image is:
+---[RSA 4096]-----+
|      o .. .      |
|o  + + E. . . o  |
|o= o + = . . o . |
|Xo= o O o . o o  |
|O+  .+.BS .  *   |
| o o o....  + .  |
|.  o o . . .    |
|      o + .      |
|      o o        |
+-----[SHA256]-----+
```

```
javillonar@wdjavillonar:~/CPE212_Javillonar$ ssh-copy-id s1javi@192.168.56.120
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
```

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 's1javi@192.168.56.120'"
and check to make sure that only the key(s) you wanted were added.

```
javillonar@wdjavillonar:~/CPE212_Javillonar$ ssh-copy-id s2javi@192.168.56.121
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
```

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 's2javi@192.168.56.121'"
and check to make sure that only the key(s) you wanted were added.

```
javillonar@wdjavillonar:~$ ssh-copy-id centjavi@192.168.56.122
The authenticity of host '192.168.56.122 (192.168.56.122)' can't be established.
ED25519 key fingerprint is SHA256:tz0qJHVVaLe8uk4pWm2brPZqCF15UA5kDt1Jo0b8Sm0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
The authenticity of host '192.168.56.122 (192.168.56.122)' can't be established.
ED25519 key fingerprint is SHA256:tz0qJHVVaLe8uk4pWm2brPZqCF15UA5kDt1Jo0b8Sm0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: 2 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
centjavi@192.168.56.122's password:
```

Number of key(s) added: 2

Now try logging into the machine, with: "ssh 'centjavi@192.168.56.122'"
and check to make sure that only the key(s) you wanted were added.

STEP 3 Create the project folder

Create the project folder and run 'mkdir -p nagios-ansible/roles/nagios/tasks' then 'cd nagios-ansible'. This sets up the folder layout: the top level holds your playbook and settings, and the roles folder holds the reusable install logic.

```
javillonar@wdjavillonar:~$ mkdir -p nagios-ansible/roles/nagios/tasks
javillonar@wdjavillonar:~$ cd nagios-ansible
javillonar@wdjavillonar:~/nagios-ansible$
```

STEP 4 Create the inventory file

Write inventory.ini This file lists your servers and groups them by OS (ubuntu, centos). Ansible reads this to know which machines exist and which group each one belongs to, so it can target them correctly

```
GNU nano 7.2 inventory.ini *
[ubuntu]
s1javi ansible_host=192.168.56.120 ansible_user=s1javi
s2javi ansible_host=192.168.56.121 ansible_user=s2javi

[centos]
centjavi ansible_host=192.168.56.122 ansible_user=centjavi

[nagios_servers:children]
ubuntu
centos
```

STEP 5 Create the Ansible config file

Write ansible.cfg and site.yml ansible.cfg just points Ansible to your inventory file. site.yml is the actual playbook — it says 'run the nagios role on every server in the nagios_servers group.'

```
javillonar@wdjavillonar:~/nagios-ansible$ nano ansible.cfg
```

```
GNU nano 7.2 ansible.cfg *
[defaults]
inventory = inventory.ini
host_key_checking = False
```

STEP 6 Create the playbook itself

Write the role's task files main.yml checks each server's OS and loads the right file. Debian.yml installs Nagios on Ubuntu (apt), RedHat.yml installs it on CentOS (yum + EPEL). This is the part that actually installs and starts Nagios.

```
javillonar@wdjavillonar:~/nagios-ansible$ nano site.yml
```

```
GNU nano 7.2 site.yml *
---
- name: Install and configure Nagios on all servers
  hosts: nagios_servers
  become: yes

  roles:
    - nagios
```

STEP 7 Create the role's entry point

Test the connection Run 'ansible ubuntu -m ping'. If it comes back with 'pong' for each server, Ansible can reach them and you're ready to run the real playbook.

```
javillonar@wdjavillonar:~/nagios-ansible$ nano roles/nagios/tasks/main.yml
```

```
GNU nano 7.2 roles/nagios/tasks/main.yml *
---
- name: Run OS-specific Nagios install tasks
  include_tasks: "[ ansible_facts['os_family'] ].yml"
```

STEP 8

Run the playbook and check it worked and run 'ansible-playbook site.yml --limit ubuntu --ask-become-pass' to install Nagios. Then run 'ansible ubuntu -a "systemctl status nagios4" --become' — seeing 'active (running)' confirms Nagios installed and started correctly.

```
javillonar@wdjavillonar:~/nagios-ansible$ nano roles/nagios/tasks/Debian.yml
```

```
GNU nano 7.2 roles/nagios/tasks/Debian.yml *
---
- name: Update apt cache
  apt:
    update_cache: yes

- name: Install Nagios4 and plugins
  apt:
    name:
      - nagios4
      - nagios-plugins-basic
    state: present

- name: Start and enable Nagios service
  service:
    name: nagios4
    state: started
    enabled: yes
```

STEP 9 Create the CentOS install steps

CentOS doesn't include Nagios in its default repositories, so the first task installs EPEL (Extra Packages for Enterprise Linux) its a repo that adds Nagios and thousands of other packages. After that, the install and service steps work the same way as the Ubuntu version.

```
javillonar@wdjavillonar:~/nagios-ansible$ nano roles/nagios/tasks/RedHat.yml
```

GNU nano 7.2

roles/nagios/tasks/RedHat.yml *

```
---
- name: Install EPEL repository
  yum:
    name: epel-release
    state: present

- name: Install Nagios and plugins
  yum:
    name:
      - nagios
      - nagios-plugins-all
    state: present

- name: Start and enable Nagios service
  service:
    name: nagios
    state: started
    enabled: yes
```

```
javillonar@wdjavillonar:~/nagios-ansible$ ls
ansible.cfg inventory.ini roles site.yml
```

STEP 11 Test that Ansible can reach all three servers

This is the Ansible's way of confirming it can SSH in and run a command. You should see "ping": "pong" s1javi,s2javi.and centjavi If it fails, go back and check step 2.

```
javillonar@wdjavillonar:~/nagios-ansible$ ansible nagios_servers -m ping
s1javi | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
s2javi | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
centjavi | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
```

STEP 12 Run the playbook on all three servers

The playbook run against every host in hosts: nagios_servers which now means all three servers. Ansible will look at each one's OS, run Debian.yml on the two Ubuntu servers and RedHat.yml on centjavi, automatically, because of how main.yml routes by OS family.

```
javillonar@wdjavillonar:~/nagios-ansible$ ansible-playbook site.yml --ask-become-pass
BECOME password:

PLAY [Install and configure Nagios on all servers] *****

TASK [Gathering Facts] *****
ok: [s2javi]
ok: [s1javi]
ok: [centjavi]

TASK [nagios : Run OS-specific Nagios install tasks] *****
included: /home/javillonar/nagios-ansible/roles/nagios/tasks/Debian.yml for s1javi, s2javi
included: /home/javillonar/nagios-ansible/roles/nagios/tasks/RedHat.yml for centjavi

TASK [nagios : Update apt cache] *****
changed: [s1javi]
changed: [s2javi]

TASK [nagios : Install Nagios4 and plugins] *****
ok: [s1javi]
ok: [s2javi]

TASK [nagios : Start and enable Nagios service] *****
ok: [s2javi]
ok: [s1javi]

TASK [nagios : Install EPEL repository] *****
ok: [centjavi]

TASK [nagios : Install Nagios and plugins] *****
ok: [centjavi]

TASK [nagios : Start and enable Nagios service] *****
ok: [s2javi]
ok: [s1javi]

TASK [nagios : Install EPEL repository] *****
ok: [centjavi]

TASK [nagios : Install Nagios and plugins] *****
ok: [centjavi]

TASK [nagios : Start and enable Nagios service] *****
ok: [centjavi]

PLAY RECAP *****
centjavi      : ok=5    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
s1javi        : ok=5    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
s2javi        : ok=5    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

STEP 13 Prove Nagios is running on all three

This runs `systemctl status nagios4` on both servers remotely and prints the result back to your terminal. Look for Active: active (running)

```
javillonar@wdjavillonar:~/nagios-ansible$ ansible ubuntu -a "systemctl status nagios4" --become -
ask-become-pass
BECOME password:
s1javi | CHANGED | rc=0 >>
● nagios4.service - nagios4
   Loaded: loaded (/usr/lib/systemd/system/nagios4.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-17 01:37:22 UTC; 17min ago
     Docs: man:nagios4
  Main PID: 29952 (nagios4)
    Tasks: 6 (limit: 3453)
  Memory: 2.8M (peak: 5.0M)
     CPU: 1.089s
   CGroup: /system.slice/nagios4.service
           └─29952 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg
           └─29954 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
           └─29955 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
           └─29956 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
           └─29957 /usr/sbin/nagios4 --worker /var/lib/nagios4/rw/nagios.qh
           └─29958 /usr/sbin/nagios4 /etc/nagios4/nagios.cfg

Sep 17 01:37:22 s1javi nagios4[29952]: wproc: Registry request: name=Core Worker 29957;pid=29957
Sep 17 01:37:22 s1javi nagios4[29952]: wproc: Registry request: name=Core Worker 29956;pid=29956
Sep 17 01:37:22 s1javi nagios4[29952]: wproc: Registry request: name=Core Worker 29955;pid=29955
Sep 17 01:37:23 s1javi nagios4[29952]: Successfully launched command file worker with pid 29958
Sep 17 01:37:23 s1javi nagios4[29952]: Successfully launched command file worker with pid 29958
Sep 17 01:41:44 s1javi nagios4[29952]: SERVICE ALERT: localhost;Swap Usage;CRITICAL;SOFT;1;SWAP C
RITICAL - 0% free (0MB out of 0MB) - Swap is either disabled, not present, or of zero size.

javillonar@wdjavillonar:~/nagios-ansible$ ansible centos -a "systemctl status nagios" --become --
ask-become-pass
BECOME password:
centjavi | CHANGED | rc=0 >>
● nagios.service - Nagios Core 4.4.14
   Loaded: loaded (/usr/lib/systemd/system/nagios.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-09-17 09:46:53 PST; 8min ago
     Docs: https://www.nagios.org/documentation
  Process: 10298 ExecStartPre=/usr/sbin/nagios -v /etc/nagios/nagios.cfg (code=exited, status=0
/SUCCESS)
  Process: 10308 ExecStart=/usr/sbin/nagios -d /etc/nagios/nagios.cfg (code=exited, status=0/SU
CESS)
  Main PID: 10318 (nagios)
    Tasks: 6 (limit: 22968)
  Memory: 25.0M (peak: 26.5M)
     CPU: 693ms
   CGroup: /system.slice/nagios.service
           └─10318 /usr/sbin/nagios -d /etc/nagios/nagios.cfg
           └─10319 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
           └─10323 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
           └─10324 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
           └─10325 /usr/sbin/nagios --worker /var/spool/nagios/cmd/nagios.qh
           └─10835 /usr/sbin/nagios -d /etc/nagios/nagios.cfg

Sep 17 09:46:53 localhost.localdomain nagios[10318]: wproc: Successfully registered manager as @w
proc with query handler
Sep 17 09:46:53 localhost.localdomain nagios[10318]: wproc: Registry request: name=Core Worker 10
```

FAILED PART:

```
javillonar@wdjavillonar:~/nagios-ansible$ ansible-playbook site.yml --ask-become-pass
BECOME password:

PLAY [Install and configure Nagios on all servers] *****

TASK [Gathering Facts] *****
ok: [centjavi]
ok: [s2javi]
ok: [s1javi]

TASK [nagios : Run OS-specific Nagios install tasks] *****
included: /home/javillonar/nagios-ansible/roles/nagios/tasks/Debian.yml for s1javi, s2javi
included: /home/javillonar/nagios-ansible/roles/nagios/tasks/RedHat.yml for centjavi

TASK [nagios : Update apt cache] *****
changed: [s2javi]
changed: [s1javi]

TASK [nagios : Install Nagios4 and plugins] *****
changed: [s2javi]
changed: [s1javi]

TASK [nagios : Start and enable Nagios service] *****
changed: [s2javi]
changed: [s1javi]

TASK [nagios : Install EPEL repository] *****
changed: [centjavi]

TASK [nagios : Install Nagios and plugins] *****
fatal: [centjavi]: FAILED! => {"changed": false, "failures": [], "msg": "Depsolve Error occurred: \n Problem: package: nagios-plugins-all-2.5-2.el9.x86_64 from epel requires nagios-plugins-disk_smb, but none of the providers can be installed\n - package nagios-plugins-disk_smb-2.5-2.el9.x86_64 from epel requires perl(utf8::all), but none of the providers can be installed\n - conflicting requests\n - nothing provides perl(PerlIO::utf8_strict) needed by perl-utf8-all-0.024-14.el9.noarch from epel\n - nothing provides perl(Import::Into) needed by perl-utf8-all-0.024-14.el9.noarch from epel", "rc": 1, "results": []}

PLAY RECAP *****
centjavi      : ok=3    changed=1    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
s1javi       : ok=5    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
s2javi       : ok=5    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

My Ubuntu control nodes and servers worked but centos not, because nagios-plugins-all tries to pull in every single plugin, including nagios-plugins-disk_smb, which needs a Perl module (perl(utf8::all)) that EPEL doesn't actually have a working package for on this CentOS version.

I fixed it swap the "install everything" package for a specific list of the plugins Nagios actually needs to function, skipping the broken one.

GNU nano 7.2 roles/nagios/tasks/RedHat.yml

```
---
- name: Install EPEL repository
  yum:
    name: epel-release
    state: present

- name: Install Nagios and plugins
  yum:
    name:
      - nagios
      - nagios-plugins-all
    state: present

- name: Start and enable Nagios service
  service:
    name: nagios
    state: started
    enabled: yes
```

I replaced it into;

GNU nano 7.2 roles/nagios/tasks/RedHat.yml *

```
---
- name: Install EPEL repository
  yum:
    name: epel-release
    state: present

- name: Install Nagios and plugins
  yum:
    name:
      - nagios
      - nagios-plugins-load
      - nagios-plugins-disk
      - nagios-plugins-users
      - nagios-plugins-procs
      - nagios-plugins-ping
      - nagios-plugins-http
    state: present

- name: Start and enable Nagios service
  service:
    name: nagios
    state: started
    enabled: yes
```

Creating new Repository:

```
javillonar@wdjavillonar:~$ sudo apt install -y git
[sudo] password for javillonar:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done

javillonar@wdjavillonar:~$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/javillonar/.git/
javillonar@wdjavillonar:~$ git add .

javillonar@wdjavillonar:~$

javillonar@wdjavillonar:~$ git config --global user.email "qmkaajavillonar@tip.edu.ph"
javillonar@wdjavillonar:~$ git config --global user.name ""
javillonar@wdjavillonar:~$ git config --global user.name ""
javillonar@wdjavillonar:~$ git config --global user.name "MadhelKathlynAnnJavillonar"
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What are the benefits of having an availability monitoring tool?

Conclusions/Learnings:

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or



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completion of the required work.

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